Ideation Phase
Define the Problem Statements

Date	04 July 2026
Team ID	
Project Name	Rising Waters: A Machine Learning Approach to Flood Prediction
Maximum Marks	2 Marks

Customer Problem Statement Template:

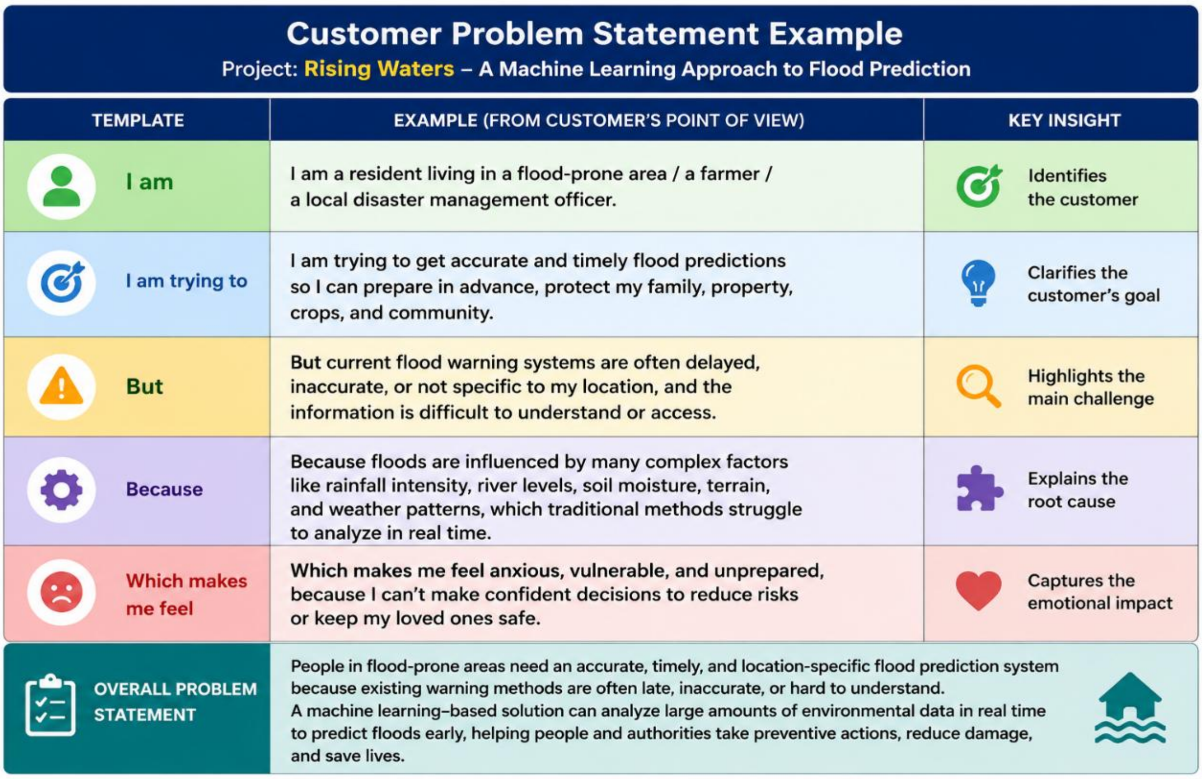
Residents living in flood-prone areas need an accurate and reliable flood prediction system that provides timely, location-specific warnings because existing flood forecasting methods often fail to deliver early and precise alerts.

A machine learning–based solution can analyse historical and real-time environmental data to improve prediction accuracy, enabling people and emergency authorities to take preventive actions, reduce damage, and save lives.

I am:	A resident living in a flood-prone area, a local disaster management officer, or a farmer whose livelihood depends on weather conditions.
I am trying to:	Prepare for potential floods by receiving accurate and timely flood predictions so I can protect my family, property, crops, and community.
But:	Current flood warning systems are often delayed, inaccurate, or difficult to access. They may not provide location-specific predictions or enough advance notice to take effective action.
Because:	Floods are influenced by multiple factors such as rainfall, river water levels, soil moisture, topography, and weather conditions. Traditional prediction methods often struggle to process these large and complex datasets in real time.
Which makes me feel:	Anxious, vulnerable, and unprepared, as I cannot confidently make decisions to minimize damage or ensure the safety of people and assets.

Reference: https://www.sciencedirect.com/journal/international-journal-of-disaster-risk-reduction

Example:



Customer Problem Statement Example

Project: Rising Waters – A Machine Learning Approach to Flood Prediction

TEMPLATE	EXAMPLE (FROM CUSTOMER'S POINT OF VIEW)	KEY INSIGHT
I am	I am a resident living in a flood-prone area / a farmer / a local disaster management officer.	Identifies the customer
I am trying to	I am trying to get accurate and timely flood predictions so I can prepare in advance, protect my family, property, crops, and community.	Clarifies the customer's goal
But	But current flood warning systems are often delayed, inaccurate, or not specific to my location, and the information is difficult to understand or access.	Highlights the main challenge
Because	Because floods are influenced by many complex factors like rainfall intensity, river levels, soil moisture, terrain, and weather patterns, which traditional methods struggle to analyze in real time.	Explains the root cause
Which makes me feel	Which makes me feel anxious, vulnerable, and unprepared, because I can't make confident decisions to reduce risks or keep my loved ones safe.	Captures the emotional impact
OVERALL PROBLEM STATEMENT	People in flood-prone areas need an accurate, timely, and location-specific flood prediction system because existing warning methods are often late, inaccurate, or hard to understand. A machine learning–based solution can analyze large amounts of environmental data in real time to predict floods early, helping people and authorities take preventive actions, reduce damage, and save lives.	

Problem Statement (PS)	I am (Customer)	I'm trying to	But	Because	Which makes me feel
PS-1	A resident living in a flood-prone area.	Receive timely flood warnings so I can protect my family and property.	Existing warning systems are often delayed or inaccurate.	Floods can occur suddenly, and current systems do not always analyze real-time environmental data effectively.	Anxious, vulnerable, and unprepared.
PS-2	A local disaster management officer.	Identify high-risk areas early and coordinate evacuations efficiently.	Available data comes from multiple sources and is difficult to analyse quickly.	Traditional forecasting methods cannot process large volumes of weather and hydrological data fast enough.	Stressed and concerned about public safety and emergency response.